

Cross-Space Distillation for One-Step Diffusion Models

CS766

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*works done at Qualcomm (with my colleagues, I didn't do this alone)

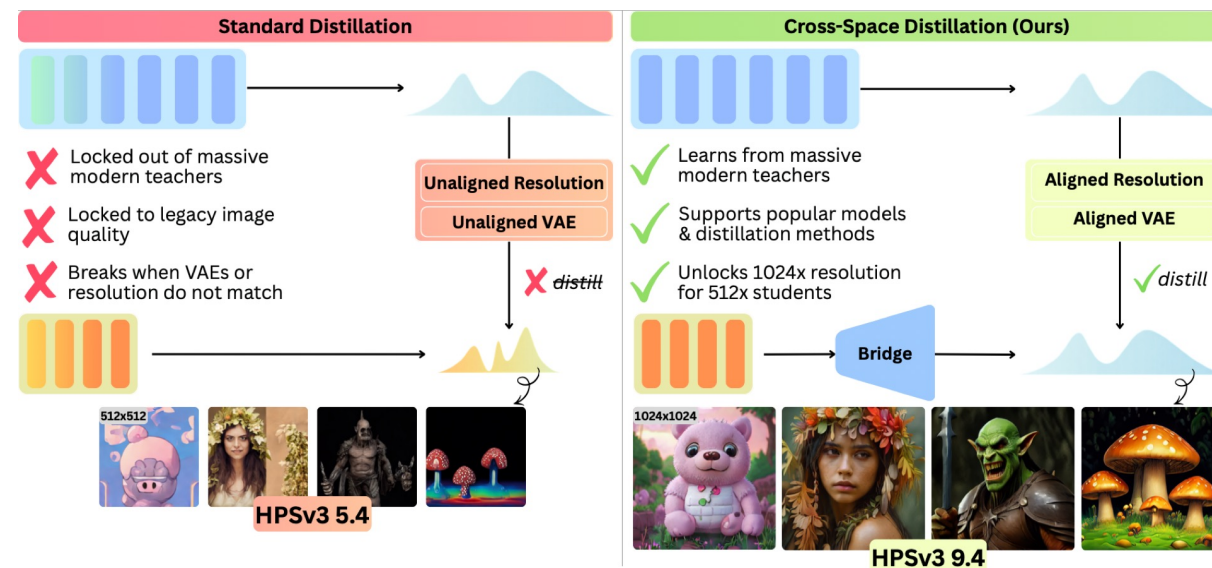
The Problem: Shared-Space Constraint

Current one-step distillation methods assume:

- **Same resolution** — both at 1024×1024
- **Same VAE** — identical image latent space

Problem: Best teachers (Flux, SD 3.5) use different resolutions and VAEs than compact students (SD 1.5)

Result: State-of-the-art quality is locked away from popular, efficient models.



Four Mismatches Between Teacher & Student

Critical Barriers (our focus)

1. Cross-Resolution — Teacher: 1024×1024, Student: 512×512

2. Cross-VAE — Different autoencoders

Manageable (empirically)

3. Cross-Architecture — DiT / MM-DiT vs UNet
Solved via output-level matching

4. Cross-Mechanism — Flow-matching vs diffusion
Solved via x_0 reparameterization

Key Insight: Resolution & VAE mismatch are the real blockers — not architecture or paradigm

Why Does This Matter?

1. Unlock the best teachers

Flux, SD 3.5 run at 1024×1024

Their quality is inaccessible to compact students today

2. Preserve ecosystems

SD 1.5 has massive ecosystem: ControlNet, LoRA, style adapters

Custom architectures break all of this

3. Practical deployment

0.86B params, single forward pass, consumer GPU compatible

Runs on mobile / edge devices

Any teacher → One student

Flux (4B, MM-DiT)

SD 3.5 (2.5B, MM-DiT)

SDXL (2.6B, UNet)

Kolors (2.6B, UNet)

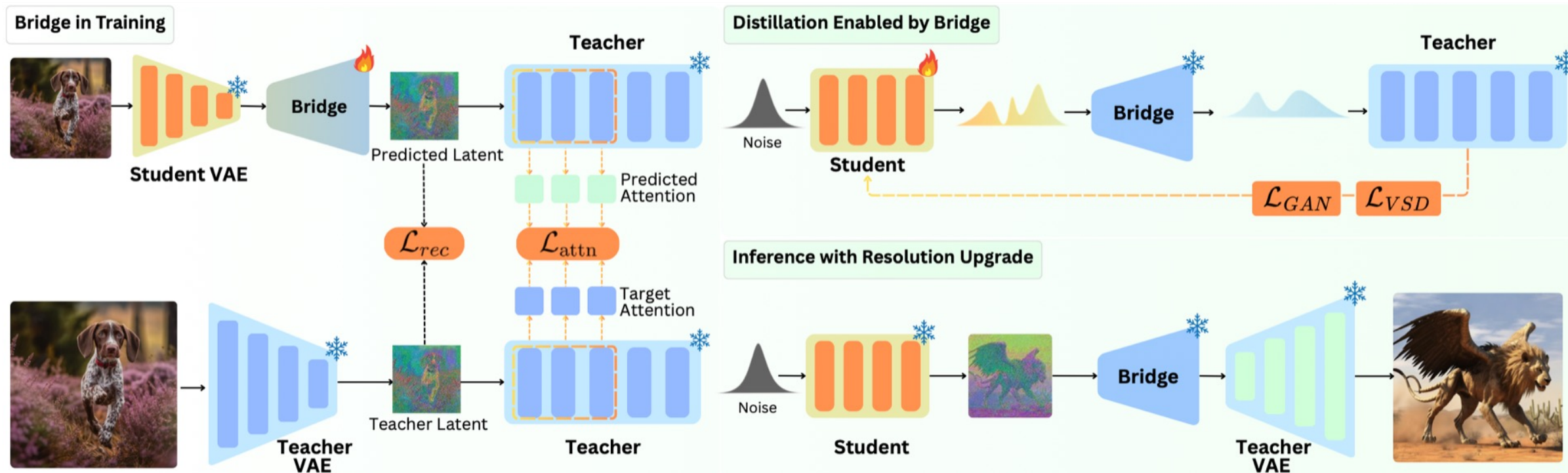
PixArt-Σ (0.6B, DiT)



SD 1.5 (0.86B) — 1 step, 512×512

Our Approach: A Lightweight Bridge

5M parameters — translates student latents into teacher-compatible latents



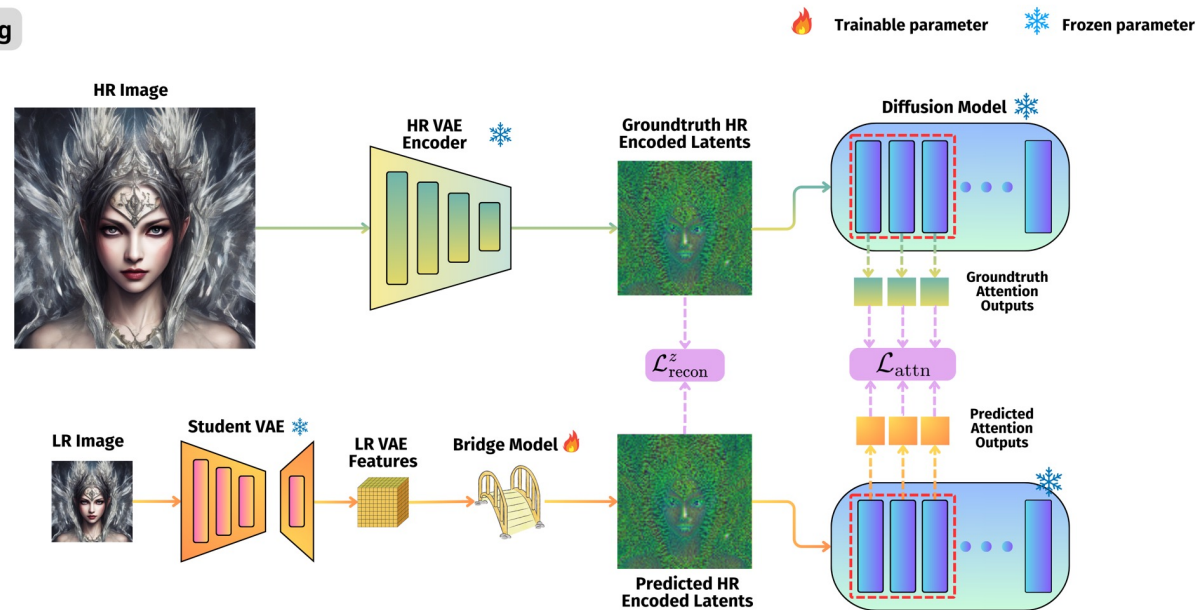
Phase 1 — Bridge Training (8h): Learn to map student latents $\rightarrow$ teacher latents using $\mathcal{L}_1$ + Attention Fidelity loss

Phase 2 — Student Distillation (20h): Frozen Bridge enables VSD + adversarial training across teacher and student

Inference: Bridge only needed during training; zero overhead at deployment. Or used for super-res

Bridge Architecture: Prior + Head

Bridge Training



Two-Stage Factorization

1. Spatial Prior (frozen)

Reuse student VAE decoder blocks

64×64 → 128×128 resolution

No learning needed — free upsampling

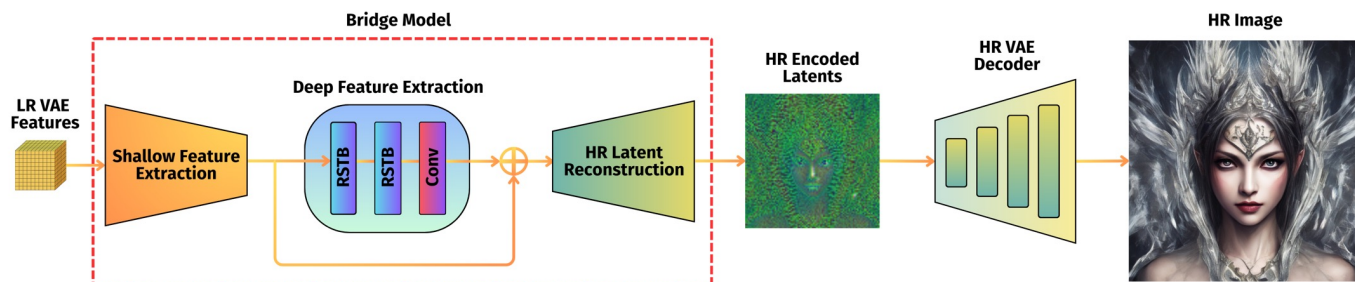
2. Projection Head (learned)

SwinIR architecture

Maps student → teacher feature space

Only 5M trainable parameters

Bridge Architecture



Key: Spatial alignment is free (frozen VAE)

Learning focuses purely on feature alignment

Training the Bridge: Two Complementary Losses

Latent Reconstruction (L_1)

$$\mathcal{L}_{rec} = \|z_T - \hat{z}_T\|_1.$$

- Direct alignment in teacher latent space
- L_1 preserves structure better than L_2
- Pixel-level supervision signal

Attention Fidelity

$$\mathcal{L}_{attn} = \sum_{l \in \text{Layers}} \tau^2 \text{KL}(P^l(\hat{z}_T; t, c) \| P^l(z_T; t, c)),$$

- Compares attention maps inside teacher
- Captures long-range semantic structure
- More stable than adversarial losses
- Reverse KL focuses on dominant modes

$$\mathcal{L}_{final} = \mathcal{L}_{rec} \text{ (pixel-level)} + \mathcal{L}_{attn} \text{ (semantic-level)}$$

Training: 2M images • 8×H100 GPUs • Only 8 hours • Then freeze Bridge for distillation

Results: SD 1.5 Student (0.86B, 1 step)

Teacher	HPSv3 ↑	HPSv2 ↑	ImgReward ↑	MPS ↑	DPG ↑
Baseline (no distill.)	5.37	21.90	-0.29	10.42	59.85
SDXL (2.6B, UNet)	9.04	27.37	0.30	11.93	63.00
Kolors (2.6B, UNet)	9.33	27.38	0.34	11.66	64.46
PixArt-Σ (0.6B, DiT)	8.65	26.43	0.30	11.94	63.43
Flux (4B, MM-DiT)	9.49	26.64	0.40	12.35	64.06
SD 3.5 Med. (2.5B)	9.42	28.30	0.62	12.96	65.75
Merged (5 teachers)	10.53	29.07	0.65	13.20	68.02

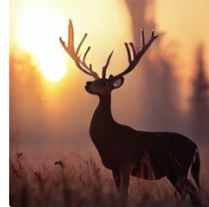
Key Improvements

HPSv3 +75%
HPSv2 +29%
ImgReward +313%

Same cost:
0.86B params
1 forward pass
512×512 output

Qualitative Results: Learning From Every Teacher

A majestic deer stands calmly in a field at sunrise with a golden sky and a softly blurred forest backdrop.



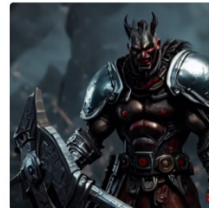
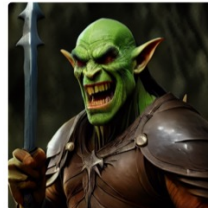
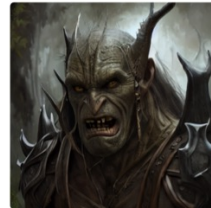
A photo of Gandalf dressed in a bright red hat with white trim and long, flowing white hair and beard, 4K.



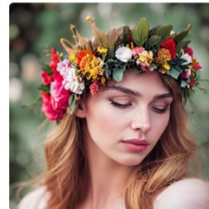
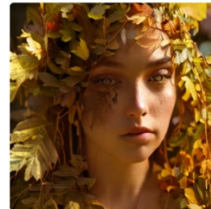
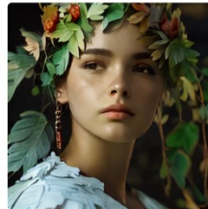
A vibrant, storybook-style castle colorful towers, and whimsical architectural in a magical fairytale world.



A dark, fantasy-styled armored orc in menacing armor with a fierce, angry gaze.



A woman with an elaborate crown made of vibrant flowers stands gracefully against a soft background.



SD 1.5 DMD2

+ SDXL

+ Kolors

+ PixArt Sigma

+ SD 3.5 Medium

+ FLUX.2-klein-4B

Ablations: Every Component Matters

Bridge Architecture

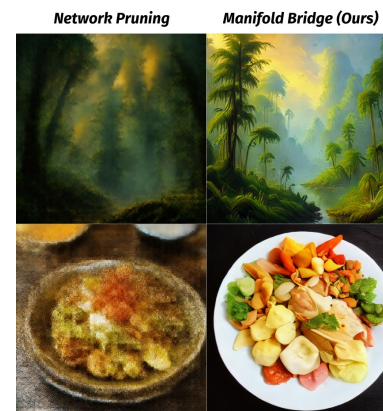
Design	L1 ↓	SSIM ↑
MLP	0.075	0.60
UNet	0.040	0.68
SwinIR	0.037	0.73
SwinIR + Spatial Prior	0.035	0.75

vs. Network Pruning

Method	Params	HPSv3
Prune 30%	1.5B	2.76
Ours	0.86B	9.42

Training Objectives

Loss	L1 ↓	SSIM ↑
L ₁ only	0.048	0.68
L ₁ + LatentLPIPS	0.042	0.71
L₁ + Attn Fidelity (ours)	0.035	0.75



Bridge Overhead:
→ SDXL: +6 ms
→ SD 3.5: +7 ms
→ Flux: +18 ms
+0.5–0.8 GB memory

Works Across Architectures & Paradigms

Teacher	Architecture	Paradigm	Latent Channels	Student HPSv3
SDXL	Large UNet	Diffusion	4	9.04
Kolors	Large UNet	Diffusion	4	9.33
PixArt-Σ	DiT	Diffusion	4	8.65
Flux	MM-DiT	Flow Matching	32	9.49
SD 3.5 Medium	MM-DiT	Flow Matching	16	9.42

Single method handles: UNet ↔ DiT ↔ MM-DiT • Diffusion ↔ Flow Matching
4 / 16 / 32 latent channels • 512 ↔ 1024 resolution
All students improve: 5.37 → 8.65–9.49 HPSv3

What We Learned

1. Resolution & VAE are the real barriers — not architecture or paradigm.

DiT→UNet and flow-matching→diffusion work fine once latent spaces align.

2. Reuse pretrained components — frozen VAE decoder blocks give spatial alignment for free.

Only 5M learnable parameters; trains in 8 hours.

3. Attention-based > adversarial supervision — Attention Fidelity loss is stable, globally aware, and hyperparameter-friendly. No training collapse risk.

4. Model merging consolidates multi-teacher knowledge — simple parameter averaging of separately distilled students yields further gains (+96% HPSv3).

Future Directions

Where this leads:

- **Video generation** — extend cross-space distillation to temporal models
- **Conditioning alignment** — handle different text encoders (CLIP vs T5) explicitly
- **Arbitrary resolutions** — beyond the 512→1024 setting
- **General latent interfaces** — Bridge as universal adapter between heterogeneous generative models

Takeaway

A 5M-parameter Bridge unlocks cross-space distillation, enabling compact models to learn from any large teacher regardless of resolution, VAE, architecture, or paradigm.

Thank You

Questions?